

Settlement & Reconciliation Control Center

System Module:	Settlement & Reconciliation	Architecture Type:	Zero-Duplication Reference Registry
Status:	PRODUCTION VERIFIED	TypeScript Validation:	Clean (0 Errors)
Date:	August 28, 2026	Multi-Currency / FX:	Directional Realized FX Engine

1. Executive Overview

The **Settlement & Reconciliation Control Center** provides enterprise-level transaction matching, multi-currency audit, and daily closing across all operational ERP modules without duplicating accounting records or requiring re-entry. It serves as an authoritative control layer connecting:

- **Cash & Roznamcha Registers** (Petty cash, counter receipts, daybooks)
- **Bank & Cheque Transactions** (Uncleared cheques, direct deposits, clearing batches)
- **Purchase & Procurement Orders** (Vendor bills, supplier advances, landed expenses)
- **Sales & Customer Accounts** (Invoices, receivables, sales recovery vouchers)
- **Payments & Inter-Branch Transfers** (Operational expenses, inter-country movements)
- **Shipping Line & Clearing Agent** (Logistics freight, agent payments, customs clearing)

2. Future-Proof Source Integration Registry

The architecture features a **Generic Event-Driven Sync Layer**. Instead of hardcoding integrations, new modules (e.g., Shipping & Clearing) seamlessly register their financial impacts directly via Postgres triggers into the Settlement engine, maintaining full decoupling and preventing duplicate accounting.

3. Database & Data Integrity Architecture

The system follows a strict **Zero Data Duplication** model. Settlement tables maintain foreign keys to authoritative ERP records:

Table / View Name	Type	Purpose
settlement_transactions	Table	Central reference registry mapping to source modules (Roznamcha, Purchase, Sales, Bank) with live remaining balances.
settlement_links	Table	Many-to-Many matching registry (1 CR to N DRs) storing permanently captured historical FX rates and realized gain/loss deltas.
settlement_audit_log	Table	Append-only, immutable audit trail recording every match, unmatched, status update, and manual flag action.
settlement_summary_v	View	Real-time daily aggregated financial summary grouped by country, branch, currency, and date.

Table / View Name	Type	Purpose
settlement_exceptions_v	View	Automated anomaly detection (unsettled >30 days, zero amounts, partial balance discrepancies).

3. REST API Subsystem

The backend exposes unified endpoints under `/app/api/erp/settlement/*` with enterprise session and scope enforcement:

Endpoint	Method	Functionality
<code>/api/erp/settlement</code>	GET	List scoped, paginated, and filtered settlement registry entries.
<code>/api/erp/settlement/dashboard</code>	GET	Aggregate KPI totals (CR, DR, remaining balances, realized FX).
<code>/api/erp/settlement/daily</code>	GET	Branch-wise daily closing reports.
<code>/api/erp/settlement/link</code>	GET / POST	Retrieve link history or execute CR→DR matching.
<code>/api/erp/settlement/link/[id]</code>	DELETE	Revert settlement link with automatic balance rollback.
<code>/api/erp/settlement/sync</code>	POST	Idempotent auto-sync engine reading new ERP transactions.
<code>/api/erp/settlement/exceptions</code>	GET	Retrieve flagged exceptions for auditor review.
<code>/api/erp/settlement/audit</code>	GET	Query immutable chronological audit history.
<code>/api/erp/settlement/fx</code>	GET	Directional multi-currency FX gain/loss analysis.
<code>/api/erp/settlement/flag</code>	POST	Toggle manual review flags on transactions.

4. Directional Multi-Currency FX Engine

Unlike conventional systems that treat rate deltas uniformly, the ERP computes FX impact based on transaction direction:

- **Credit (CR) Receipts:** If settlement rate yields higher USD than original $\$ \rightarrow \$$ **FX Realized Gain (+)**. If lower $\$ \rightarrow \$$ **FX Realized Loss (-)**.
- **Debit (DR) Disbursements:** If settlement rate requires fewer USD to settle than original $\$ \rightarrow \$$ **FX Realized Gain (+)**. If more $\$ \rightarrow \$$ **FX Realized Loss (-)**.
- **Rate Preservation:** Historical conversion rates are permanently locked in `settlement_links` for lifelong compliance.

5. UI & Navigation Integration

The module is integrated directly into the ERP sidebar navigation (SETTLEMENT & RECONCILIATION) across 13 dedicated routes:

- **Settlement Dashboard** (`/dashboard/settlement`)
- **Daily Settlement & Closing** (`/dashboard/settlement/daily`)
- **Cash / Roznamcha Settlement** (`/dashboard/settlement/cash`)
- **Bank & Cheque Settlement** (`/dashboard/settlement/bank`)
- **Party & Account Reconciliation** (`/dashboard/settlement/party`)

- **Purchase Bill Settlement** (/dashboard/settlement/purchase)
- **Sales & Customer Settlement** (/dashboard/settlement/sales)
- **Payment & Transfer Settlement** (/dashboard/settlement/payment)
- **Expense & Office Bill Settlement** (/dashboard/settlement/expense)
- **Multi-Currency / FX Center** (/dashboard/settlement/fx)
- **Unsettled & Partial Queue** (/dashboard/settlement/unsettled)
- **Universal Reports Hub** (/dashboard/settlement/reports)
- **Settlement Audit Trail** (/dashboard/settlement/audit)

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